Lab 7: MOSFETS and JFETS Op Amps

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This lab investigates the behaviors of MOSFETS and JFETS in various circuits. Field-effect transistors (FETs) are semiconductor devices in which current through a channel is controlled by an electric field applied at the gate. Unlike BJTs, which are current-controlled devices, FETs operate by using voltage to modulate conductivity, resulting in extremely high input impedance and low power consumption. These experiments investigate two major Metal-Oxide-Semiconductor FETs (MOSFETs) and Junction FETs (JFETs). Each relies on controlling the flow of charge through a semiconductor channel using electric fields.

A MOSFET consists of a source, drain, and gate terminal. Between the source and drain lies a doped semiconductor region. Above this channel is a thin insulating oxide, typically SiO_2, and on top a conductive gate material that is usually silicon or a metal. Since the gate is insulated, virtually no current flows into it. When a gate-to-source voltage is applied an electric field forms an inversion layer at the semiconductor-oxide interface. This creates a conductive channel that allows carriers, electrons in an n-channel MOSFET or holes in a p-channel MOSFET, to flow. This makes MOSFETs ideal for switching and amplification, especially in digital logic circuitry where minimal gate current is essential.

A JFET, oppositely uses a reverse-biased pn-junction at the gate to control the channel. It alters conduction by expanding or narrowing the depletion region created at the junction. JFETs are typically depletion-mode devices, meaning the channel exists at zero bias and is cut off as gate voltage becomes more negative (for n-channel devices).

These devices are widely used in analog and digital circuits: MOSFETs are typically seen in integrated circuits such as processors and memory due to their scalability and low power operation, while JFETs usually operate in low-noise analog and op-amp circuits due to their low gate leakage and great noise performance.

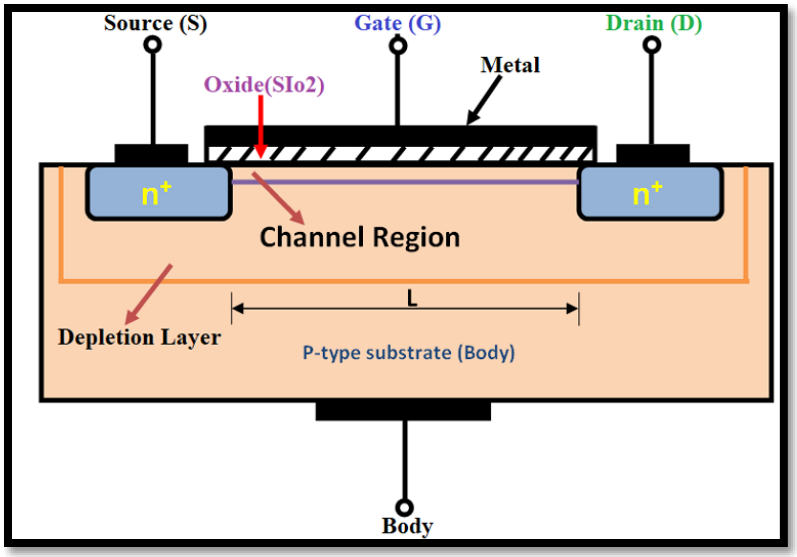

Figure 1: MOSFET Structure

Fundamentals:

A Junction Field-Effect Transistor (JFET) controls current using a reverse-biased pn-junction instead of an oxide-isolated gate. An n-channel JFET consists of an n-type semiconductor bar forming the channel, with p-type gate regions diffused on opposite sides. When there is no gate-to-source voltage the channel will allow the drain current to flow. If a negative voltage acts on the gate it reverse-biases the pn-junction. This widens the depletion region and narrows the channel. As the depletion regions expand inward the drain current will decrease as well. When the depletion regions touch it cuts off and the drain current saturates at a nearly constant value even if drain-source voltage increases.

Compared to BJTs, JFETs offer much higher input impedance since the gate current is limited to reverse-bias leakage. They also generate less noise, making them suitable for precision analog circuits. However, they do provide less gain and their individual device-to-device properties vary more since manufacturing can be complex. Compared to MOSFETs, JFETs are more robust against static electricity and are more resistant to oxide breakdown. But MOSFETs are far more scalable and are seen far more often. Especially in all modern integrated circuits.

JFETs are more suited for low-noise amplifiers or high-impedance buffer stages such as the OP482 used in this lab. Their depletion-mode operation and low gate leakage make them reliable for precision analog applications that require stable low-noise performance.

<u>Experiment 1: Demonstrating the Transfer Characteristics of a MOSFET</u>

The goal of experiment 1 is to measure the characteristics of a MOSFET. A sweep is run for V_{GS} and V_{DS} and the data is recorded and plotted. The transconductance g_m is found as a function of V_{GS}. Python is used to generate the plots. The data is fit to the square law theory for FETs to further observe the device's behavior. The data is as follows:

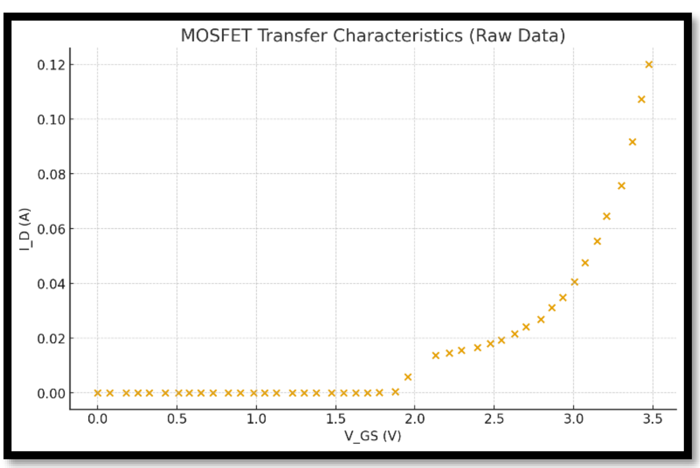

Figure 2: MOSFET V_{GS} vs I_D Raw Data

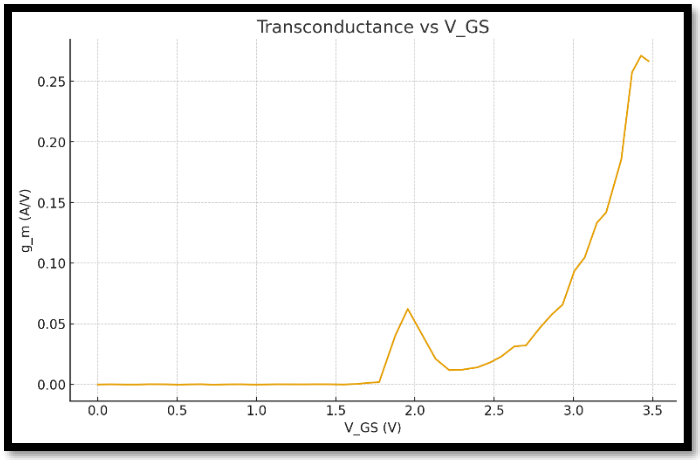

Figure 3: Transconductance g_m as Function of V_{GS}

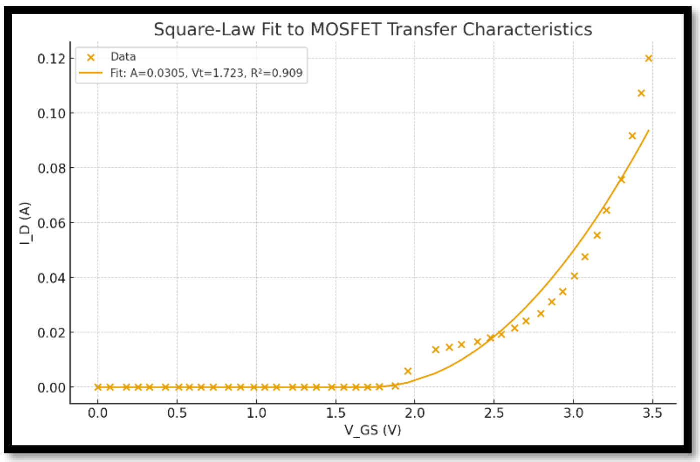

Figure 4: Transconductance g_m as Function of V_{GS}

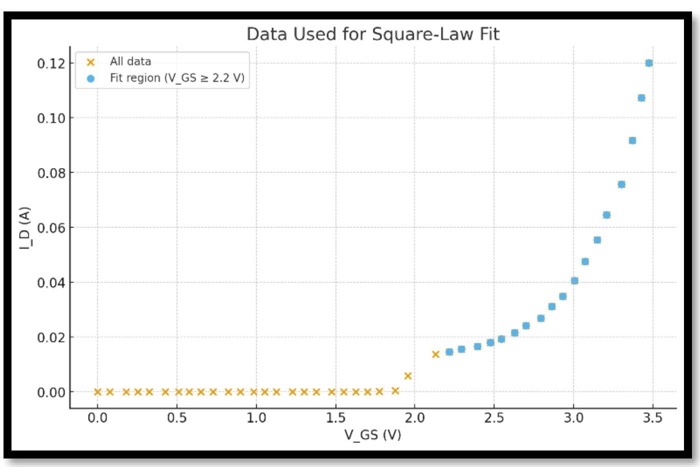

Figure 5: Data Used to help fit to Square Law

The measured MOSFET transfer characteristics show the expected behavior of an N-channel device. In this case, the 5LN01SP MOSFET. At low gate-source voltages $V_{GS} < 2V$, the drain current remains nearly zero. This corresponds to the cutoff region where the electric field induced at the gate isn't enough to create a conductive n-channel. Once V_{GS} exceeds about 2 V, the drain current begins to rise rapidly and eventually follows the characteristic quadratic predicted by square-law MOSFET theory:

$$I_D = A(V_{GS} - V_T)^2$$

This behavior matches the theoretical operation of a MOSFET in saturation (as shown in the lecture notes). An increased gate voltage increases the channel's charge density leading to a the nonlinear rise in drain current.

Fitting the upper region of the curve, where Square Law would apply, produced a threshold voltage of approximately $V_T \approx 1.72$ V and a quadratic factor $A \approx 0.0305$ as can be seen in the graph. The threshold voltage is a little higher than what the datasheet suggests but this discrepancy is somewhat expected because the value is highly dependent on the chosen current level and measurement setup. The square-law fit achieved an R^2 value of approximately 0.91, indicating reasonably good agreement with theory, especially considering the simplifications inherent in the ideal MOSFET model.

Several factors explain the deviations from perfect quadratic behavior. First, real MOSFETs exhibit channel-length modulation, causing $I_D$to increase slightly with V_{DS}, even in saturation. This is what actually distorts that ideal parabolic shape. Furthermore, at lower gate voltages the device is most likely operating beneath its threshold. This is where the drain current rises exponentially rather than quadratically. It explains the small nonzero measured currents below the threshold region. At higher drain currents, heat and series resistances can also cause discrepancies from the ideal model.

Overall, the transfer curve matches the expected MOSFET theory. There is a clear threshold behavior, a transition into strong inversion, and a rising current that fits the quadratic law in its appropriate region. The few deviations are consistent with real-world device physics and measurement limitations, supporting that the experiment successfully demonstrates MOSFET transfer characteristics and validates the underlying square-law model.

Concept Question(s):

The recorded data fits well with the square law theory. The value $V_T \approx 1.72$ V is a bit high compared to the data sheet, but it did not cause any issues. This is most likely due to a less ideal measurement setup and experimenting conditions.

Experiment 2: Demonstrating the use of an Op Amp

The goal of experiment 2 is to build various circuits to demonstrate the behavior of an OP27 Op Amp chip in noninverting, inverting, and summer conditions. Resistor values and voltage inputs vary. The output waveform of three circuits is recorded and analyzed to calculate the gain. The general breadboard setup is shown in the figure below:

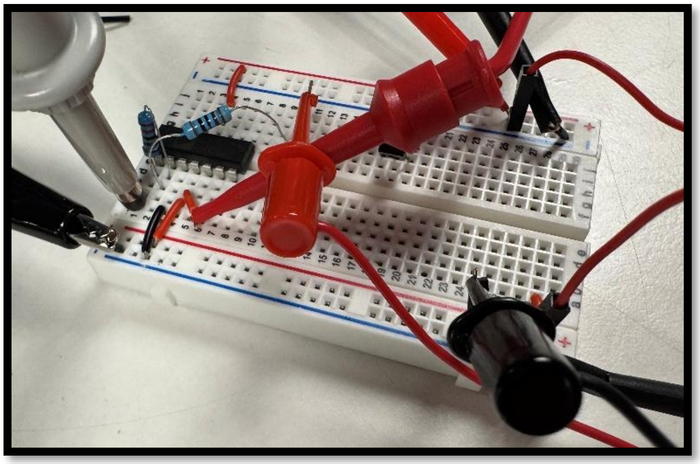

Figure 6: General Experiment 2 Setup

Non-inverting circuit: R1 = 1kΩ, R2 = 1kΩ, Sin wave input: Freq: 500 Hz, Amplitude: 2 Vpp

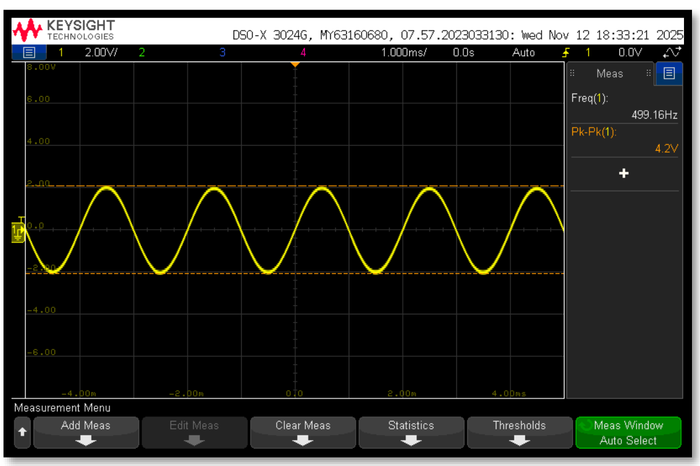

Figure 7: Noninverting Output

Inverting circuit: R1 = 1kΩ, R2 = 2.2kΩ, Sin wave input: Freq: 1 kHz, Amplitude: 2 Vpp

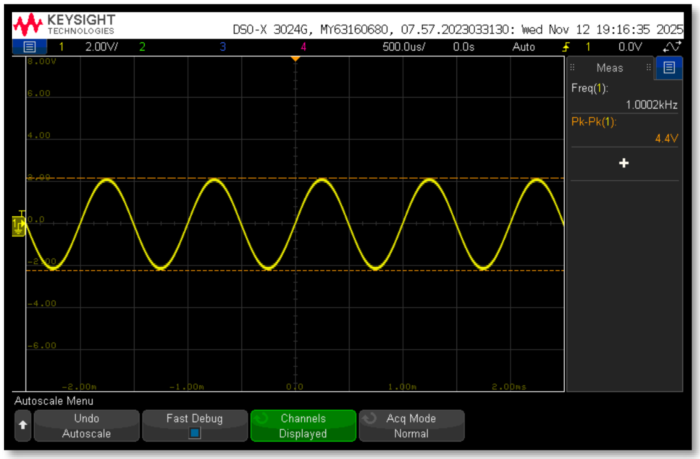

Figure 8: Inverting Output

Summing circuit: R1 = 1k, R2 = 1.5k, RF = 2.2k, V1 5V, V2 5V

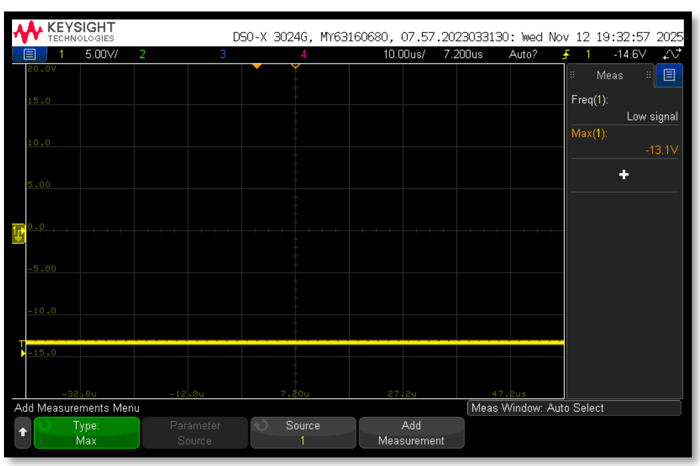

Figure 9: Summer Output

Non-inverting Output Analysis:

$$K = 1 + \frac{R2}{R1}$$

$$v_o = \left(1 + \frac{R2}{R1}\right) \cdot v_s$$

$$K = 1 + \frac{1k\Omega}{1k\Omega} = 2$$

The oscilloscope image with a pk-pk value of 4.2V agrees with this calculation. The slight 0.2V deviation is likely due to resistor tolerances and the general breadboard setup not being perfectly ideal.

Inverting Output Analysis:

$$K = \frac{-R2}{R1}$$

$$v_o = -\left(\frac{R2}{R1}\right) \cdot v_s$$

$$K = \frac{-2.2k\Omega}{1k\Omega} = -2.2$$

The oscilloscope image with a pk-pk value of 4.4V supports this calculation. It lines up exactly with what is expected. The output waveform is clean and unclipped indicating the OP27 is functioning correctly. The inversion can clearly be seen in the measurement as well.

Summer Output Analysis:

$$v_o = -R_F\left(\frac{v_1}{R_1}+\frac{v_2}{R_2}\right)$$

$$v_o = -2.2k\Omega\left(\frac{5V}{1k\Omega}+\frac{5V}{1.5k\Omega}\right) = -18.3\,V$$

With the utilized resistor values, the theoretical output calculated is -18.3 V. However, the Op Amp is powered by ±15 V so it is impossible to actually achieve that value. The oscilloscope shows an output of -13.1 V which is a more realistic value given the limitations of the OP27. The circuit tries to output the full -18.3 V but rails out at -13.1 V creating that flat line on the oscilloscope. This shows that op amps can't produce an infinite output voltage even if the theoretical gain equation gives a larger value.

In my measurements, I only recorded oscilloscope data at a single frequency for each circuit (500 Hz for the noninverting amplifier and 1 kHz for the inverting amplifier). I did not run a frequency sweep on one of the circuits to be able to plot the roll-off in gain. At these lower frequencies, the op amp operates within its flat-gain region. Meaning the observed gain should closely match the ideal closed-loop gains determined by the resistor ratios. According to the datasheet, the OP27 op amp has a gain-bandwidth product of about 8 MHz.

For a closed-loop gain of approximately 2, the expected −3 dB bandwidth is about

$$f_c \approx 8\text{ MHz}/2 \approx 4\text{ MHz}$$

which a lot higher than the 500 Hz–1 kHz range used in this experiment. Because my measurements are so far below the expected cutoff frequency, they do not show any noticeable roll-off in gain.

This means, with my current data, I am unable to show the roll-off in gain that might happen at larger frequencies. However, it can be explained relatively easily. Theoretically, I can use the relationship between gain-bandwidth products and closed-loop bandwidth to explain what would is expected. The gain should remain constant at low frequencies, then begin to fall at a rate of approximately −20 dB/decade after the cutoff frequency. The absence of a measured plot and a clear roll-off region make it impossible to determine exactly what it would be for my non-inverting setup though.

Concept Question(s):

Overall, the non-inverting, inverting, and summing amplifier all behaved relatively as expected compared to the op-amp theory. The oscilloscope measurements largely support the expected gain. The only discrepancy is with the experimental result of the summing amplifier compared to the theoretical. But this is simply due to the op amp's limitation.

Experiment 3: Using an Op Amp in a Transimpedance Amplifier

The goal of experiment 3 is to use an op amp in a transimpedance amplifier to convert current into voltage without the losses acquired if instead using a basic resistor. The following transimpedance amplifier circuit is built and an SMU is used a current source. The DC current-to-voltage is measured and a gain is calculated. It is done again but with a photodiode instead to generate the current. The R_F value used is 10 kΩ.

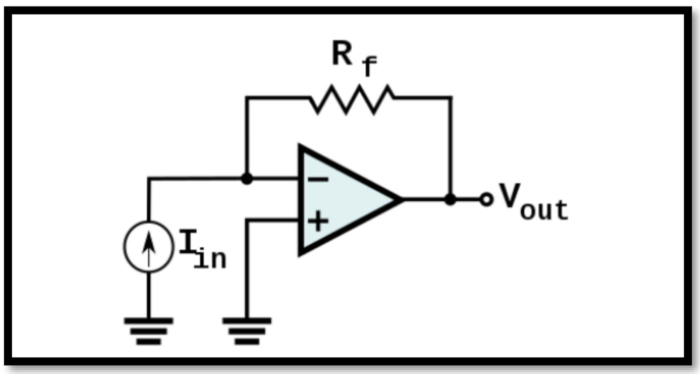

Figure 10: Transimpedance Amplifier Circuit

Using SMU as current source:

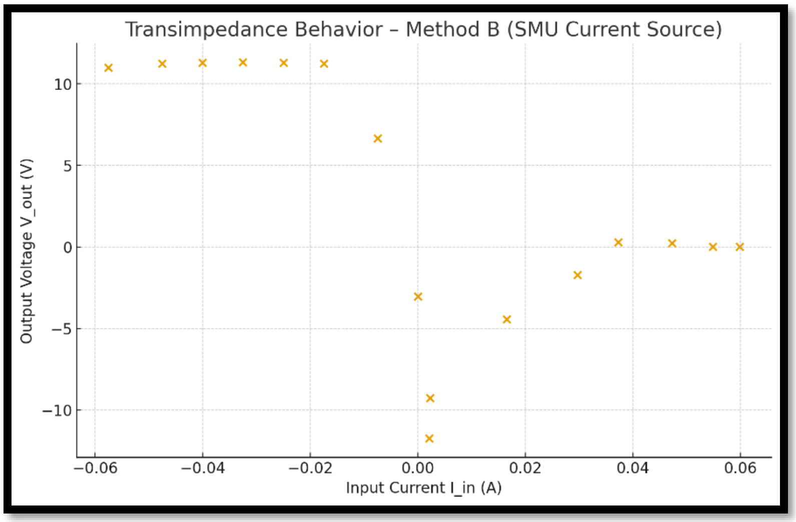

Figure 11: SMU as Current Source Data

It is clear in the beginning few points the op amp output is just about constant at around 11 V even as the input current changes. It then dives to the negatives and returns to about 0 V. This curve illustrates the TIA behaves approximately linearly but only over a limited range of input currents. It saturates near the supply rails so transimpedance largely varies with current.

$$R_{\{TIA\}} \approx \frac{V_{out}}{I_{in}}$$

Using that formula with the first few points a value of 360 Ω is found for R_{TIA}. This means that for about very 1ma of input current, the output voltage will change by:

$$V_{out} = (0.001\ A)(360\ \Omega) = 0.36\ V$$

This value is relatively consistent with the slope seen in the linear region of the plot.

Using photodiode as current source:

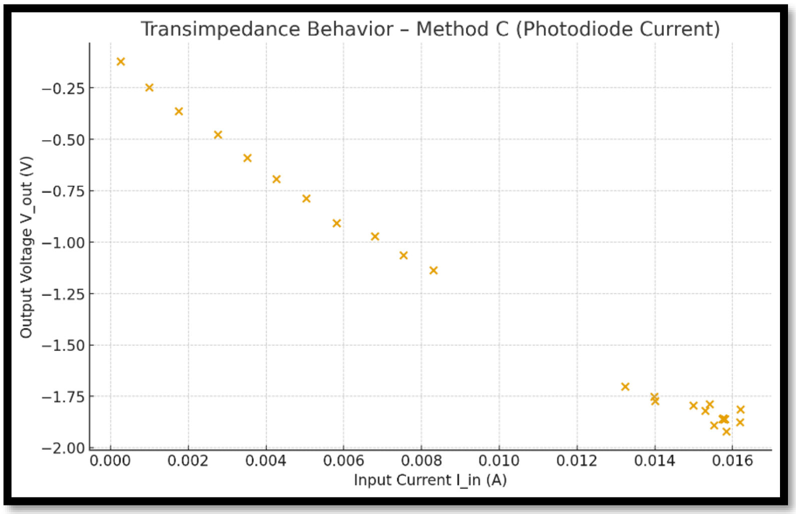

Figure 12: SMU as Current Source Data

Over the mid/high-current region (roughly I_{in} ≈ 5–16 mA) the points lie close to a straight line, which matches a TIA with approximately constant negative gain.

$$R_{\{TIA\}} \approx \frac{V_{out}}{I_{in}}$$

Using that formula with the those mid/high-current points a value of 118Ω is found for R_{TIA}. This means that for about very 1ma of input current, the output voltage will change by:

$$V_{out} = (0.001\ A)(118\ \Omega) = 0.118\ V$$

This value is relatively consistent with the slope seen through the whole plot.

<u>Concept Question(s):</u>

Overall, the transimpedance amplifier did not behave as ideal as expected. There were significant differences between methods B and C. Method B utilized the SMU as a current source and exhibited significant deviation from ideal behavior. The amplifier saturated at both positive and negative supply rails as the SMU forced current in both directions, producing extreme voltages (e.g., +11 V, –12 V). This saturation corrupted the measurement and caused the apparent transimpedance to vary wildly (from ~200 Ω to almost 900 Ω depending on the points used). A large amount of the data lies outside the linear operating region of the amplifier, and therefore the gain calculated from Method B is not representative of the true circuit behavior. These deviations can largely be explained by the SMU forcing current beyond the safe linear region of the op-amp, rail-to-rail saturation (which destroys the linear V = IR relationship), and sign reversals with noise near zero crossing.

However, using the photodiode to produce a current, a much cleaner dataset is achieved for Method C. This is clearly seen in the plot. The photodiode current stayed well within the linear operating region of the op-amp, and no saturation occurred. As a result, the transimpedance gain was consistent and matched theoretical behavior. Deviations were small and attributable to the photodiode dark current and noise at very low current levels and small op-amp input bias currents.

Conclusion:

Across all three experiments, the behavior of MOSFETs, JFET-based op-amps, and transimpedance amplifiers closely matched theoretical expectations. These experiments served to demonstrate how voltage-controlled devices shape modern analog circuits. In Experiment 1, the MOSFET produced a clear quadratic transfer curve, and the threshold voltage of ~1.72 V supported the square-law model despite minor deviations. Experiment 2 confirmed textbook op-amp gains. For example, the non-inverting amplifier achieved a gain of ~2, consistent with $1 + R_2/R_1$. Experiment 3 showed that the TIA accurately converted current to voltage, with Method C producing a stable transimpedance of 118 Ω. Overall, I learned how device physics translates into measurable circuit behavior and how measurement factors like noise and saturation can influence data in unexpected ways.

GenAI Usage:

AI was used to help generate figures 11 and 12 from the recorded .csv data file.

Citations:

MOSFET Structure Image:

Venn, Matt. “MOSFET.” *Zero to ASIC Course*, 12 Nov. 2020, www.zerotoasiccourse.com/terminology/mosfet/